

Solution	Marks	
SECTION A 1. (a) Let $P(n)$ be the proposition $\sum_{k=1}^n \frac{k(k-1)}{2} = \frac{(n-1)n(n+1)}{6}$. For $n = 1$: L.H.S. = $\frac{1(1-1)}{2} = 0$ R.H.S. = $\frac{(1-1)(1)(1+1)}{6} = 0 = \text{L.H.S.}$ $\therefore P(1)$ is true. Assume $P(m)$ is true for some positive integer m, i.e., $\sum_{k=1}^m \frac{k(k-1)}{2} = \frac{(m-1)m(m+1)}{6}$ For $n = m + 1$: L.H.S. = $\sum_{k=1}^{m+1} \frac{k(k-1)}{2} = \sum_{k=1}^m \frac{k(k-1)}{2} + \frac{(m+1)m}{2}$ $= \frac{(m-1)m(m+1)}{6} + \frac{m(m+1)}{2}$ $= \frac{m(m+1)}{6} [(m-1) + 3]$ $= \frac{m(m+1)(m+2)}{6}$ $= \frac{[(m+1)-1](m+1)[(m+1)+1]}{6} = \text{R.H.S.}$ $\therefore P(m+1)$ is true if $P(m)$ is true. By mathematical induction, $P(n)$ is true for all positive integers n. (b) $\sum_{k=1}^{10} \frac{k(k-1)}{2} = \frac{(10-1)(10)(10+1)}{6}$ $= \frac{9 \times 10 \times 11}{6} = 165$	1M 1M 1M 1A 1M 1A	

Solution	Marks	
2. (a) Let $P(n)$ be the proposition $\sum_{i=1}^n \frac{1}{i^2 + i} = \frac{n}{n+1}$. For $n = 1$: L.H.S. $= \frac{1}{1^2 + 1} = \frac{1}{2}$ R.H.S. $= \frac{1}{1+1} = \frac{1}{2} = \text{L.H.S.}$ $\therefore P(1)$ is true. Assume $P(k)$ is true for some positive integer k, i.e., $\sum_{i=1}^k \frac{1}{i^2 + i} = \frac{k}{k+1}$ For $n = k + 1$: $\begin{aligned} \text{L.H.S.} &= \sum_{i=1}^{k+1} \frac{1}{i^2 + i} = \sum_{i=1}^k \frac{1}{i^2 + i} + \frac{1}{(k+1)^2 + (k+1)} \\ &= \frac{k}{k+1} + \frac{1}{(k+1)(k+2)} \\ &= \frac{k(k+2) + 1}{(k+1)(k+2)} = \frac{k^2 + 2k + 1}{(k+1)(k+2)} \\ &= \frac{(k+1)^2}{(k+1)(k+2)} = \frac{k+1}{(k+1)+1} = \text{R.H.S.} \end{aligned}$ $\therefore P(k+1)$ is true if $P(k)$ is true. By mathematical induction, $P(n)$ is true for all positive integers n. (b) $\sum_{i=1}^{99} \frac{1}{i^2 + i} = \frac{99}{99+1}$ $= \frac{99}{100}$	1M 1M 1M 1A 1M 1A	

Solution	Marks	
3. (a) Let $P(n)$ be the proposition $a_n = 2^{n+1} + n - 2$. For $n = 1$: $a_1 = 2^1 + 1 = 3$ $\text{R.H.S.} = 2^{1+1} + 1 - 2 = 3 = a_1 = \text{L.H.S.}$ $\therefore P(1)$ is true. Assume $P(k)$ is true for some positive integer k, i.e., $a_k = 2^{k+1} + k - 2$ For $n = k + 1$: $\begin{aligned} a_{k+1} &= a_k + (2^{k+1} + 1) \\ &= (2^{k+1} + k - 2) + 2^{k+1} + 1 \\ &= 2 \cdot 2^{k+1} + k - 1 \\ &= 2^{(k+1)+1} + (k + 1) - 2 = \text{R.H.S.} \end{aligned}$ $\therefore P(k + 1)$ is true if $P(k)$ is true. By mathematical induction, $P(n)$ is true for all positive integers n. (b) Let $Q(n)$ be the proposition $b_n = \frac{1}{3} \left(2^{n+2} + 6n - 24 + \frac{19}{2^{n-1}} \right)$. For $n = 1$: $b_1 = 3$ $\text{R.H.S.} = \frac{1}{3} \left(2^3 + 6(1) - 24 + \frac{19}{2^0} \right) = \frac{1}{3}(9) = 3 = b_1$ $\therefore Q(1)$ is true. Assume $Q(k)$ is true for some positive integer k, i.e., $b_k = \frac{1}{3} \left(2^{k+2} + 6k - 24 + \frac{19}{2^{k-1}} \right)$ For $n = k + 1$:	1M 1M 1M 1A 1M 1M	

Solution	Marks	
$ \begin{aligned} b_{k+1} &= \frac{1}{2}b_k + a_k \\ &= \frac{1}{2} \cdot \frac{1}{3} \left(2^{k+2} + 6k - 24 + \frac{19}{2^{k-1}} \right) + (2^{k+1} + k - 2) \\ &= \frac{1}{3} \left(2^{k+1} + 3k - 12 + \frac{19}{2^k} \right) + \frac{1}{3}(3 \cdot 2^{k+1} + 3k - 6) \\ &= \frac{1}{3} \left(4 \cdot 2^{k+1} + 6k - 18 + \frac{19}{2^k} \right) \\ &= \frac{1}{3} \left(2^{(k+1)+2} + 6(k+1) - 24 + \frac{19}{2^{(k+1)-1}} \right) = \text{R.H.S.} \end{aligned} $ $\therefore Q(k+1)$ is true if $Q(k)$ is true. By mathematical induction, $Q(n)$ is true for all positive integers n.	1M	
	1A	

Solution	Marks	
4. (a) Let $P(n)$ be the proposition $\sum_{r=1}^n \frac{r}{r^4 + r^2 + 1} = \frac{n(n+1)}{2(n^2 + n + 1)}$. For $n = 1$: $\text{L.H.S.} = \frac{1}{1^4 + 1^2 + 1} = \frac{1}{3}$ $\text{R.H.S.} = \frac{1(1+1)}{2(1^2 + 1 + 1)} = \frac{2}{6} = \frac{1}{3} = \text{L.H.S.}$ $\therefore P(1)$ is true. Assume $P(k)$ is true for some positive integer k, i.e., $\sum_{r=1}^k \frac{r}{r^4 + r^2 + 1} = \frac{k(k+1)}{2(k^2 + k + 1)}$ For $n = k + 1$: $\begin{aligned} \text{L.H.S.} &= \sum_{r=1}^{k+1} \frac{r}{r^4 + r^2 + 1} \\ &= \frac{k(k+1)}{2(k^2 + k + 1)} + \frac{k+1}{(k+1)^4 + (k+1)^2 + 1} \\ &= \frac{k+1}{2} \left[\frac{k}{k^2 + k + 1} + \frac{2}{(k^2 + k + 1)(k^2 + 3k + 3)} \right] \\ &= \frac{k+1}{2(k^2 + k + 1)} \cdot \frac{k(k^2 + 3k + 3) + 2}{k^2 + 3k + 3} \\ &= \frac{k+1}{2(k^2 + k + 1)} \cdot \frac{(k^2 + k + 1)(k+2)}{k^2 + 3k + 3} \\ &= \frac{(k+1)(k+2)}{2[(k+1)^2 + (k+1) + 1]} = \text{R.H.S.} \end{aligned}$ $\therefore P(k+1)$ is true if $P(k)$ is true. By mathematical induction, $P(n)$ is true for all positive integers n. (b) $\begin{aligned} \sum_{r=33}^{64} \frac{r}{r^4 + r^2 + 1} &= \sum_{r=1}^{64} \frac{r}{r^4 + r^2 + 1} - \sum_{r=1}^{32} \frac{r}{r^4 + r^2 + 1} \\ &= \frac{64 \times 65}{2(64^2 + 64 + 1)} - \frac{32 \times 33}{2(32^2 + 32 + 1)} \\ &= \frac{2080}{4161} - \frac{528}{1057} \\ &= \frac{1552}{4398177} \end{aligned}$	1M 1M 1M 1A 1M 1A	

Solution	Marks	
5. (a) Note that $\left(\frac{1}{2}\right)^{-i} = 2^i$. Let $P(n)$ be the proposition $\sum_{i=1}^n 2^i \cdot i^3 = 2^{n+1}(n^3 - 3n^2 + 9n - 13) + 26$. For $n = 1$: L.H.S. $= 2^1 \cdot 1^3 = 2$ R.H.S. $= 2^2(1 - 3 + 9 - 13) + 26 = 4(-6) + 26 = 2 = \text{L.H.S.}$ $\therefore P(1)$ is true. Assume $P(k)$ is true for some positive integer k, i.e., $\sum_{i=1}^k 2^i \cdot i^3 = 2^{k+1}(k^3 - 3k^2 + 9k - 13) + 26$ For $n = k + 1$: L.H.S. $= \sum_{i=1}^{k+1} 2^i \cdot i^3 = 2^{k+1}(k^3 - 3k^2 + 9k - 13) + 26 + 2^{k+1}(k+1)^3$ $= 2^{k+1} [(k^3 - 3k^2 + 9k - 13) + (k^3 + 3k^2 + 3k + 1)] + 26$ $= 2^{k+1}(2k^3 + 12k - 12) + 26 = 2^{k+2}(k^3 + 6k - 6) + 26$ $= 2^{(k+1)+1} [(k+1)^3 - 3(k+1)^2 + 9(k+1) - 13] + 26 = \text{R.H.S.}$ $\therefore P(k+1)$ is true if $P(k)$ is true. By mathematical induction, $P(n)$ is true for all positive integers n. (b) $\sum_{i=8}^{13} \left(\frac{1}{2}\right)^{-i} \cdot i^3 = \sum_{i=1}^{13} 2^i \cdot i^3 - \sum_{i=1}^7 2^i \cdot i^3$ $= [2^{14}(13^3 - 3(13^2) + 9(13) - 13) + 26]$ $- [2^8(7^3 - 3(7^2) + 9(7) - 13) + 26]$ $= [16384(1794) + 26] - [256(246) + 26]$ $= 29392896 - 62976$ $= 29329920$	1M 1M 1M 1A 1M 1A	

Solution	Marks	
6. (a) Let $P(n)$ be the proposition $\sum_{i=1}^n \frac{3i^4 + i^6}{(-1)^i} = \frac{(-1)^n n^3 (n+1)^3}{2}$. For $n = 1$: $\text{L.H.S.} = \frac{3(1)^4 + (1)^6}{(-1)^1} = -4$ $\text{R.H.S.} = \frac{(-1)^1 (1)^3 (1+1)^3}{2} = \frac{-8}{2} = -4 = \text{L.H.S.}$ $\therefore P(1)$ is true. Assume $P(k)$ is true for some positive integer k, i.e., $\sum_{i=1}^k \frac{3i^4 + i^6}{(-1)^i} = \frac{(-1)^k k^3 (k+1)^3}{2}$ For $n = k + 1$: $\begin{aligned} \text{L.H.S.} &= \sum_{i=1}^{k+1} \frac{3i^4 + i^6}{(-1)^i} = \sum_{i=1}^k \frac{3i^4 + i^6}{(-1)^i} + \frac{3(k+1)^4 + (k+1)^6}{(-1)^{k+1}} \\ &= \frac{(-1)^k k^3 (k+1)^3}{2} + (-1)^{k+1} (k+1)^4 [3 + (k+1)^2] \\ &= \frac{(-1)^{k+1} (k+1)^3}{2} [-k^3 + 2(k+1)(k^2 + 2k + 4)] \\ &= \frac{(-1)^{k+1} (k+1)^3}{2} (k^3 + 6k^2 + 12k + 8) \\ &= \frac{(-1)^{k+1} (k+1)^3 (k+2)^3}{2} \\ &= \frac{(-1)^{k+1} (k+1)^3 [(k+1) + 1]^3}{2} = \text{R.H.S.} \end{aligned}$ $\therefore P(k+1)$ is true if $P(k)$ is true. By mathematical induction, $P(n)$ is true for all positive integers n. (b) $\sum_{i=9}^{16} \frac{3i^4 + i^6}{(-1)^{i+1}} = - \sum_{i=9}^{16} \frac{3i^4 + i^6}{(-1)^i} = - \left(\sum_{i=1}^{16} \frac{3i^4 + i^6}{(-1)^i} - \sum_{i=1}^8 \frac{3i^4 + i^6}{(-1)^i} \right)$ $\begin{aligned} &= - \left[\frac{(-1)^{16} (16)^3 (17)^3}{2} - \frac{(-1)^8 (8)^3 (9)^3}{2} \right] \\ &= - \left(\frac{4096 \times 4913}{2} - \frac{512 \times 729}{2} \right) \\ &= -(10061824 - 186624) \\ &= -9875200 \end{aligned}$	1M 1M 1M 1A 1M 1A	

Solution	Marks	
7. (a) Let $P(k)$ be the proposition $\sum_{n=1}^k \frac{1}{n(n+1)(n+2)} = \frac{k(k+3)}{4(k+1)(k+2)}$. For $k = 1$: $\text{L.H.S.} = \frac{1}{1(2)(3)} = \frac{1}{6}$ $\text{R.H.S.} = \frac{1(4)}{4(2)(3)} = \frac{1}{6} = \text{L.H.S.}$ $\therefore P(1)$ is true. Assume $P(m)$ is true for some positive integer m, i.e., $\sum_{n=1}^m \frac{1}{n(n+1)(n+2)} = \frac{m(m+3)}{4(m+1)(m+2)}$ For $k = m + 1$: $\begin{aligned} \text{L.H.S.} &= \sum_{n=1}^{m+1} \frac{1}{n(n+1)(n+2)} \\ &= \frac{m(m+3)}{4(m+1)(m+2)} + \frac{1}{(m+1)(m+2)(m+3)} \\ &= \frac{m(m+3)^2 + 4}{4(m+1)(m+2)(m+3)} = \frac{m^3 + 6m^2 + 9m + 4}{4(m+1)(m+2)(m+3)} \\ &= \frac{(m+1)^2(m+4)}{4(m+1)(m+2)(m+3)} = \frac{(m+1)[(m+1)+3]}{4[(m+1)+1][(m+1)+2]} = \text{R.H.S.} \end{aligned}$ $\therefore P(m+1)$ is true if $P(m)$ is true. By mathematical induction, $P(k)$ is true for all positive integers k. (b) Let $Q(k)$ be the proposition $\sum_{n=1}^k \frac{2n+k}{n(n+1)(n+2)} = \frac{k+4}{4} - \frac{5k+4}{2(k+1)(k+2)}$. For $k = 1$: $\text{L.H.S.} = \frac{2(1)+1}{1(2)(3)} = \frac{1}{2}$ $\text{R.H.S.} = \frac{1+4}{4} - \frac{5(1)+4}{2(2)(3)} = \frac{1}{2} = \text{L.H.S.}$ $\therefore Q(1)$ is true. Assume $Q(m)$ is true for some positive integer m, i.e., $\sum_{n=1}^m \frac{2n+m}{n(n+1)(n+2)} = \frac{m+4}{4} - \frac{5m+4}{2(m+1)(m+2)}$ For $k = m + 1$:	1M 1M 1M 1A 1M 1M	

Solution	Marks	
L.H.S. = $\sum_{n=1}^{m+1} \frac{2n + (m+1)}{n(n+1)(n+2)}$ = $\sum_{n=1}^m \frac{2n + (m+1)}{n(n+1)(n+2)} + \frac{2(m+1) + (m+1)}{(m+1)(m+2)(m+3)}$ = $\sum_{n=1}^m \frac{2n + m}{n(n+1)(n+2)} + \sum_{n=1}^m \frac{1}{n(n+1)(n+2)} + \frac{2(m+1) + (m+1)}{(m+1)(m+2)(m+3)}$ = $\frac{m+4}{4} - \frac{5m+4}{2(m+1)(m+2)} + \frac{m(m+3)}{4(m+1)(m+2)} + \frac{2(m+1) + (m+1)}{(m+1)(m+2)(m+3)}$ = $\frac{m+4}{4} + \frac{m(m+3) - 2(5m+4)}{4(m+1)(m+2)} + \frac{3(m+1)}{(m+1)(m+2)(m+3)}$ = $\frac{m+4}{4} + \frac{m^2 - 7m - 8}{4(m+1)(m+2)} + \frac{3}{(m+2)(m+3)}$ = $\frac{m+4}{4} + \frac{(m+1)(m-8)}{4(m+1)(m+2)} + \frac{3}{(m+2)(m+3)}$ = $\frac{m+4}{4} + \frac{m-8}{4(m+2)} + \frac{3}{(m+2)(m+3)}$ = $\frac{(m+4)(m+2) + (m-8)}{4(m+2)} + \frac{3}{(m+2)(m+3)}$ = $\frac{m^2 + 7m}{4(m+2)} + \frac{3}{(m+2)(m+3)}$ = $\frac{(m^2 + 7m)(m+3) + 12}{4(m+2)(m+3)}$ = $\frac{m^3 + 10m^2 + 21m + 12}{4(m+2)(m+3)}$ = $\frac{(m+5)(m+2)(m+3) - 2(5m+9)}{4(m+2)(m+3)}$ = $\frac{m+5}{4} - \frac{5m+9}{2(m+2)(m+3)}$ = $\frac{(m+1) + 4}{4} - \frac{5(m+1) + 4}{2[(m+1) + 1][(m+1) + 2]} = \text{R.H.S.}$ $\therefore Q(m+1)$ is true if $Q(m)$ is true. By mathematical induction, $Q(k)$ is true for all positive integers k. (c) Let $m = n + 1$. When $n = 6$, $m = 7$; when $n = 18$, $m = 19$.	1M 1A	

Solution	Marks	
$ \begin{aligned} & \sum_{n=6}^{18} \frac{2n+21}{(n+1)(n+2)(n+3)} \\ &= \sum_{m=7}^{19} \frac{2(m-1)+21}{m(m+1)(m+2)} \\ &= \sum_{m=7}^{19} \frac{2m+19}{m(m+1)(m+2)} \\ &= \sum_{m=1}^{19} \frac{2m+19}{m(m+1)(m+2)} - \sum_{m=1}^6 \frac{2m+19}{m(m+1)(m+2)} \\ &= \sum_{m=1}^{19} \frac{2m+19}{m(m+1)(m+2)} - \sum_{m=1}^6 \frac{2m+6}{m(m+1)(m+2)} - \sum_{m=1}^6 \frac{13}{m(m+1)(m+2)} \\ &= \frac{19+4}{4} - \frac{5(19)+4}{2(19+1)(19+2)} - \left(\frac{6+4}{4} - \frac{5(6)+4}{2(6+1)(6+2)} \right) - 13 \left(\frac{6(6+3)}{4(6+1)(6+2)} \right) \\ &= \frac{169}{560} \end{aligned} $	1M 1A	

Solution	Marks	
8. (a) Let $P(n)$ be the proposition. $\sum_{k=1}^n (\frac{1+\sqrt{5}}{2})^{k-1} [k(\frac{1+\sqrt{5}}{2}) - (k-1)] = n\phi^n$. For $n = 1$: L.H.S. $= (\frac{1+\sqrt{5}}{2})^0 [1(\frac{1+\sqrt{5}}{2}) - 0] = \frac{1+\sqrt{5}}{2}$ R.H.S. $= (1)(\frac{1+\sqrt{5}}{2})^1 = \frac{1+\sqrt{5}}{2} = \text{L.H.S.}$ $\therefore P(1)$ is true. Assume $P(m)$ is true for some positive integer m, i.e., $\sum_{k=1}^m (\frac{1+\sqrt{5}}{2})^{k-1} [k(\frac{1+\sqrt{5}}{2}) - (k-1)] = m(\frac{1+\sqrt{5}}{2})^m$. For $n = m + 1$: L.H.S. $= \sum_{k=1}^{m+1} (\frac{1+\sqrt{5}}{2})^{k-1} [k(\frac{1+\sqrt{5}}{2}) - (k-1)]$ $= \sum_{k=1}^m (\frac{1+\sqrt{5}}{2})^{k-1} [k(\frac{1+\sqrt{5}}{2}) - (k-1)] + (\frac{1+\sqrt{5}}{2})^m [(m+1)(\frac{1+\sqrt{5}}{2}) - (m+1-1)]$ $= m(\frac{1+\sqrt{5}}{2})^m + (\frac{1+\sqrt{5}}{2})^m [(m+1)(\frac{1+\sqrt{5}}{2}) - (m+1-1)]$ $= (\frac{1+\sqrt{5}}{2})^m [m + (m+1)(\frac{1+\sqrt{5}}{2}) - m]$ $= (m+1)(\frac{1+\sqrt{5}}{2})^{m+1}$ $\therefore P(m+1)$ is true if $P(m)$ is true. By mathematical induction, $P(n)$ is true for all positive integers n. (b) $\sum_{k=1}^{10^{10}} (\frac{1+\sqrt{5}}{2})^{(10^{10})} - (\frac{1+\sqrt{5}}{2})^{k-3} [(\frac{1+\sqrt{5}}{2}) + 1] [k(\frac{1+\sqrt{5}}{2}) - (k-1)]$ $= \sum_{k=1}^{10^{10}} (\frac{1+\sqrt{5}}{2})^{(10^{10})} - \sum_{k=1}^{10^{10}} \frac{(\frac{1+\sqrt{5}}{2})^{k-1}}{(\frac{1+\sqrt{5}}{2})^2} [(\frac{1+\sqrt{5}}{2}) + 1] [k(\frac{1+\sqrt{5}}{2}) - (k-1)]$ $= (10^{10})(\frac{1+\sqrt{5}}{2})^{(10^{10})} - \sum_{k=1}^{10^{10}} \frac{(\frac{1+\sqrt{5}}{2})^{k-1}}{(\frac{3+\sqrt{5}}{2})} (\frac{3+\sqrt{5}}{2}) [k(\frac{1+\sqrt{5}}{2}) - (k-1)]$ $= (10^{10})(\frac{1+\sqrt{5}}{2})^{(10^{10})} - \sum_{k=1}^{10^{10}} (\frac{1+\sqrt{5}}{2})^{k-1} [k(\frac{1+\sqrt{5}}{2}) - (k-1)]$ $= (10^{10})(\frac{1+\sqrt{5}}{2})^{(10^{10})} - (10^{10})(\frac{1+\sqrt{5}}{2})^{(10^{10})}$ $= 0$	1M 1M 1M 1A 1M 1A	

Solution	Marks	
9. (a) $u_2 = 3u_1 - 2 = 3(2) - 2 = 4$ $u_3 = 3u_2 - 2 = 3(4) - 2 = 10$ (b) Let $P(n)$ be the proposition $u_n = 3^{n-1} + 1$. For $n = 1$: R.H.S. $= 3^{1-1} + 1 = 1 + 1 = 2 = \text{L.H.S.}$ $\therefore P(1)$ is true. Assume $P(k)$ and $P(k+1)$ are true for some positive integer k, i.e., $u_k = 3^{k-1} + 1$ For $n = k + 1$: $\begin{aligned} u_{k+1} &= 3u_k - 2 \\ &= 3(3^{k-1} + 1) - 2 \\ &= 3^k + 3 - 2 \\ &= 3^k + 1 = \text{R.H.S.} \end{aligned}$ $\therefore P(k+1)$ is true if $P(k)$ is true. By mathematical induction, $P(n)$ is true for all positive integers n. (c)(i) $S_n = \sum_{r=1}^n (3^{r-1} + 1) = \sum_{r=1}^n 3^{r-1} + \sum_{r=1}^n 1 = \frac{3^n - 1}{3 - 1} + n$ $= \frac{3^n + 2n - 1}{2}$ (c)(ii) Let $Q(n)$ be the proposition that $3^n + 2n - 1$ is divisible by 2. For $n = 1$: $3^1 + 2(1) - 1 = 4$, which is divisible by 2. $\therefore Q(1)$ is true. Assume $Q(k)$ is true for some positive integer k, i.e., $3^k + 2k - 1 = 2M$ for some integer M. For $n = k + 1$: $\begin{aligned} 3^{k+1} + 2(k+1) - 1 &= 3(3^k) + 2k + 2 - 1 \\ &= 3(3^k) + 2k + 1 \\ &= 3(3^k + 2k - 1) - 3(2k - 1) + 2k + 1 \\ &= 3(2M) - 6k + 3 + 2k + 1 \\ &= 6M - 4k + 4 \\ &= 2(3M - 2k + 2) \end{aligned}$ Since $3M - 2k + 1$ is an integer, $3^{k+1} + 2(k+1) - 1$ is divisible by 2. $\therefore Q(k+1)$ is true if $Q(k)$ is true.	1A 1A 1M 1M 1M 1A 1A 1M 1M 1M	

Solution	Marks	
By mathematical induction, $Q(n)$ is true for all positive integers n .	1A	

Solution	Marks	
SECTION B 10. (a) Let $P(n)$ be the proposition $\sum_{k=1}^n \frac{1}{(k+1)\sqrt{k} + k\sqrt{k+1}} = 1 - \frac{1}{\sqrt{n+1}}$. For $n = 1$: L.H.S. = $\frac{1}{2\sqrt{1} + 1\sqrt{2}} = \frac{1}{2 + \sqrt{2}} = \frac{2 - \sqrt{2}}{2} = 1 - \frac{1}{\sqrt{2}}$ R.H.S. = $1 - \frac{1}{\sqrt{1+1}} = 1 - \frac{1}{\sqrt{2}} = \text{L.H.S.}$ $\therefore P(1)$ is true. Assume $P(m)$ is true for some positive integer m, i.e., $\sum_{k=1}^m \frac{1}{(k+1)\sqrt{k} + k\sqrt{k+1}} = 1 - \frac{1}{\sqrt{m+1}}$ For $n = m + 1$: L.H.S. = $\sum_{k=1}^{m+1} \frac{1}{(k+1)\sqrt{k} + k\sqrt{k+1}}$ $= 1 - \frac{1}{\sqrt{m+1}} + \frac{1}{(m+2)\sqrt{m+1} + (m+1)\sqrt{m+2}}$ $= 1 - \frac{1}{\sqrt{m+1}} + \frac{(m+2)\sqrt{m+1} - (m+1)\sqrt{m+2}}{(m+1)(m+2)[(m+2) - (m+1)]}$ $= 1 - \frac{1}{\sqrt{m+1}} + \frac{\sqrt{m+1}}{m+1} - \frac{\sqrt{m+2}}{m+2}$ $= 1 - \frac{1}{\sqrt{m+1}} + \frac{1}{\sqrt{m+1}} - \frac{1}{\sqrt{m+2}}$ $= 1 - \frac{1}{\sqrt{m+2}} = 1 - \frac{1}{\sqrt{(m+1)+1}} = \text{R.H.S.}$ $\therefore P(m+1)$ is true if $P(m)$ is true. By mathematical induction, $P(n)$ is true for all positive integers n. (b) $\sum_{k=1225}^{2024} \frac{1}{(k+1)\sqrt{k} + k\sqrt{k+1}} = \sum_{k=1}^{2024} \frac{1}{(k+1)\sqrt{k} + k\sqrt{k+1}} - \sum_{k=1}^{1224} \frac{1}{(k+1)\sqrt{k} + k\sqrt{k+1}}$ $= \left(1 - \frac{1}{\sqrt{2025}}\right) - \left(1 - \frac{1}{\sqrt{1225}}\right)$ $= \left(1 - \frac{1}{45}\right) - \left(1 - \frac{1}{35}\right) = \frac{1}{35} - \frac{1}{45}$ $= \frac{2}{315}$	1M	
	1M	
	1M	
	1M	
	1A	
	1M	
	1A	

Solution	Marks	
11. (a) Let $P(n)$ be the proposition $\sum_{k=1}^n \tan k\theta \tan(k-1)\theta = \frac{\tan n\theta}{\tan \theta} - n$. For $n = 1$: L.H.S. $= \tan \theta \tan 0 = 0$ R.H.S. $= \frac{\tan \theta}{\tan \theta} - 1 = 0 = \text{L.H.S.}$ $\therefore P(1)$ is true. Assume $P(m)$ is true for some positive integer m, i.e., $\sum_{k=1}^m \tan k\theta \tan(k-1)\theta = \frac{\tan m\theta}{\tan \theta} - m$ For $n = m + 1$: L.H.S. $= \sum_{k=1}^{m+1} \tan k\theta \tan(k-1)\theta$ $= \left(\frac{\tan m\theta}{\tan \theta} - m \right) + (\tan(m+1)\theta \tan m\theta)$ $= \frac{\tan m\theta}{\tan \theta} + \tan(m+1)\theta \tan m\theta - m$ $= \tan m\theta \left(\frac{1}{\tan \theta} + \tan(m+1)\theta \right) - m$ $= \tan m\theta \left(\frac{\cos \theta}{\sin \theta} + \frac{\sin(m+1)\theta}{\cos(m+1)\theta} \right) - m$ $= \tan m\theta \left(\frac{\cos \theta \cos(m+1)\theta + \sin(m+1)\theta \sin \theta}{\sin \theta \cos(m+1)\theta} \right) - m$ $= \tan m\theta \left(\frac{\frac{\cos(1+m+1)\theta + \cos[1-(m+1)]\theta}{2} + \frac{\cos[1-(m+1)]\theta - \cos(1+m+1)\theta}{2}}{\sin \theta \cos(m+1)\theta} \right) - m$ $= \tan m\theta \left(\frac{\cos(m+2)\theta + \cos m\theta + \cos m\theta - \cos(m+2)\theta}{2 \sin \theta \cos(m+1)\theta} \right) - m$ $= \frac{\sin m\theta}{\sin \theta \cos(m+1)\theta} - m$ $= \frac{\sin(m+1-1)\theta}{\sin \theta \cos(m+1)\theta} - m$ $= \frac{\sin(m+1)\theta \cos \theta - \cos(m+1)\theta \sin \theta}{\sin \theta \cos(m+1)\theta} - m$ $= \frac{\tan(m+1)\theta}{\tan \theta} - (m+1) = \text{R.H.S.}$ $\therefore P(m+1)$ is true if $P(m)$ is true. By mathematical induction, $P(n)$ is true for all positive integers n. (b) Let $j = k + 6$. When $k = 1$, $j = 7$; when $k = 6$, $j = 12$.	1M 1M 1M 1A	

Solution	Marks	
$\sum_{k=1}^6 \tan(k+6)\theta \tan(k+5)\theta = \sum_{j=7}^{12} \tan j\theta \tan(j-1)\theta$ $= \sum_{j=1}^{12} \tan j\theta \tan(j-1)\theta - \sum_{j=1}^6 \tan j\theta \tan(j-1)\theta$ $= \left(\frac{\tan 12\theta}{\tan \theta} - 12 \right) - \left(\frac{\tan 6\theta}{\tan \theta} - 6 \right)$ $= \frac{\tan 12\theta - \tan 6\theta}{\tan \theta} - 6$ Given $\frac{\tan 12\theta - \tan 6\theta}{\tan \theta} - 6 = -6 \implies \tan 12\theta = \tan 6\theta.$ $12\theta - 6\theta = \pi$ or $12\theta - 6\theta = 2\pi$ for $0 < \theta < \frac{\pi}{2}$ Solving each equation: $12\theta - 6\theta = \pi \implies 6\theta = \pi \implies \theta = \frac{\pi}{6}$ $12\theta - 6\theta = 2\pi \implies 6\theta = 2\pi \implies \theta = \frac{\pi}{3}$ $\therefore$ we have $\theta = \frac{\pi}{6}$ or $\frac{\pi}{3}$.	1M 1M 1A	

Solution	Marks	
12. (a) Let $P(n)$ be the proposition $\sum_{k=1}^n \frac{\sin 2k\theta}{\cos(2k-1)\theta \cos(2k+1)\theta} = \frac{\sec(2n+1)\theta - \sec \theta}{2 \sin \theta}$. For $n = 1$: $\text{R.H.S.} = \frac{\frac{1}{\cos 3\theta} - \frac{1}{\cos \theta}}{2 \sin \theta} = \frac{\cos \theta - \cos 3\theta}{2 \sin \theta \cos \theta \cos 3\theta} = \frac{2 \sin 2\theta \sin \theta}{2 \sin \theta \cos \theta \cos 3\theta} = \frac{\sin 2\theta}{\cos \theta \cos 3\theta} = \text{L.H.S.}$ $\therefore P(1)$ is true. Assume $P(m)$ is true for some positive integer m, i.e., $\sum_{k=1}^m \frac{\sin 2k\theta}{\cos(2k-1)\theta \cos(2k+1)\theta} = \frac{\sec(2m+1)\theta - \sec \theta}{2 \sin \theta}$ For $n = m + 1$: $\begin{aligned} \text{L.H.S.} &= \sum_{k=1}^{m+1} \frac{\sin 2k\theta}{\cos(2k-1)\theta \cos(2k+1)\theta} \\ &= \frac{\sec(2m+1)\theta - \sec \theta}{2 \sin \theta} + \frac{\sin 2(m+1)\theta}{\cos(2m+1)\theta \cos(2m+3)\theta} \\ &= \frac{1}{2 \sin \theta} \left[\sec(2m+1)\theta - \sec \theta + \frac{2 \sin \theta \sin(2m+2)\theta}{\cos(2m+1)\theta \cos(2m+3)\theta} \right] \\ &= \frac{1}{2 \sin \theta} \left[\sec(2m+1)\theta - \sec \theta + \frac{\cos(2m+1)\theta - \cos(2m+3)\theta}{\cos(2m+1)\theta \cos(2m+3)\theta} \right] \\ &= \frac{1}{2 \sin \theta} [\sec(2m+1)\theta - \sec \theta + \sec(2m+3)\theta - \sec(2m+1)\theta] \\ &= \frac{\sec(2m+3)\theta - \sec \theta}{2 \sin \theta} = \frac{\sec[2(m+1)+1]\theta - \sec \theta}{2 \sin \theta} = \text{R.H.S.} \end{aligned}$ $\therefore P(m+1)$ is true if $P(m)$ is true. By mathematical induction, $P(n)$ is true for all positive integers n. (b) Set $\theta = \frac{\pi}{16}$ and $n = 256$ in the identity from (a): $\begin{aligned} \sum_{k=1}^{256} \frac{\sin \frac{k\pi}{8}}{\cos \frac{(2k-1)\pi}{16} \cos \frac{(2k+1)\pi}{16}} &= \frac{\sec \left(513 \cdot \frac{\pi}{16} \right) - \sec \frac{\pi}{16}}{2 \sin \frac{\pi}{16}} \\ &= \frac{\sec \left(32\pi + \frac{\pi}{16} \right) - \sec \frac{\pi}{16}}{2 \sin \frac{\pi}{16}} \\ &= \frac{\sec \frac{\pi}{16} - \sec \frac{\pi}{16}}{2 \sin \frac{\pi}{16}} \\ &= 0 \end{aligned}$	1M 1M 1M 1A 1M 1A	

Solution	Marks	
13. (a) $B^2 = \begin{pmatrix} 0 & 3 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 3 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$ (b) Let $P(n)$ be the proposition $A^n = 4^n I + n4^{n-1}B$. For $n = 1$: L.H.S. $= A^1 = 4I + B$ R.H.S. $= 4^1 I + (1)4^0 B = 4I + B = \text{L.H.S.}$ $\therefore P(1)$ is true. Assume $P(m)$ is true for some positive integer m, i.e., $A^m = 4^m I + m4^{m-1}B$ For $n = m + 1$: L.H.S. $= A^{m+1} = A^m A$ $= (4^m I + m4^{m-1}B)(4I + B)$ $= 4^{m+1}I + 4^m B + m4^m B + m4^{m-1}B^2$ $= 4^{m+1}I + (m+1)4^m B + \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \quad (\text{since } B^2 = \mathbf{0})$ $= 4^{m+1}I + (m+1)4^{(m+1)-1}B = \text{R.H.S.}$ $\therefore P(m+1)$ is true if $P(m)$ is true. By mathematical induction, $P(n)$ is true for all positive integers n.	1A 1M 1M 1M 1A	

Solution	Marks	
14. Let $P(n)$ be the proposition $A^n = \begin{pmatrix} 2^n & 0 & 0 \\ n2^n & 2^n & 0 \\ n(n-1)2^{n-1} & n2^n & 2^n \end{pmatrix}$. For $n = 1$: $\text{R.H.S.} = \begin{pmatrix} 2^1 & 0 & 0 \\ (1)2^1 & 2^1 & 0 \\ (1)(0)2^0 & (1)2^1 & 2^1 \end{pmatrix} = \begin{pmatrix} 2 & 0 & 0 \\ 2 & 2 & 0 \\ 0 & 2 & 2 \end{pmatrix} = \text{L.H.S.}$ $\therefore P(1)$ is true. Assume $P(m)$ is true for some positive integer m, i.e., $A^m = \begin{pmatrix} 2^m & 0 & 0 \\ m2^m & 2^m & 0 \\ m(m-1)2^{m-1} & m2^m & 2^m \end{pmatrix}$ For $n = m + 1$: L.H.S. $= A^{m+1} = A^m A$ $= \begin{pmatrix} 2^m & 0 & 0 \\ m2^m & 2^m & 0 \\ m(m-1)2^{m-1} & m2^m & 2^m \end{pmatrix} \begin{pmatrix} 2 & 0 & 0 \\ 2 & 2 & 0 \\ 0 & 2 & 2 \end{pmatrix}$ $= \begin{pmatrix} 2^{m+1} & 0 & 0 \\ 2 \cdot m2^m + 2 \cdot 2^m & 2^{m+1} & 0 \\ 2m(m-1)2^{m-1} + 2m2^m & 2m2^m + 2 \cdot 2^m & 2^{m+1} \end{pmatrix}$ $= \begin{pmatrix} 2^{m+1} & 0 & 0 \\ (m+1)2^{m+1} & 2^{m+1} & 0 \\ m(m-1)2^m + m2^{m+1} & (m+1)2^{m+1} & 2^{m+1} \end{pmatrix}$ $= \begin{pmatrix} 2^{m+1} & 0 & 0 \\ (m+1)2^{m+1} & 2^{m+1} & 0 \\ m(m+1)2^m & (m+1)2^{m+1} & 2^{m+1} \end{pmatrix} = \text{R.H.S.}$	1M 1M 1M	

Solution	Marks	
$\therefore P(m+1)$ is true if $P(m)$ is true. By mathematical induction, $P(n)$ is true for all positive integers n .	1A	

Solution	Marks	
15. (a) (i) $\mathbf{v}_1 = \mathbf{i} + 2\mathbf{j}$, $\mathbf{v}_2 = 2\mathbf{i} + 3\mathbf{j}$. $\mathbf{v}_1 \cdot \mathbf{v}_2 = (1)(2) + (2)(3) = 2 + 6 = 8$ (ii) $P_1 = \mathbf{v}_1 \cdot \mathbf{v}_2 = 8$ $\mathbf{v}_3 = 3\mathbf{i} + 4\mathbf{j} \implies \mathbf{v}_2 \cdot \mathbf{v}_3 = (2)(3) + (3)(4) = 6 + 12 = 18.$ $P_2 = (\mathbf{v}_1 \cdot \mathbf{v}_2) + (\mathbf{v}_2 \cdot \mathbf{v}_3) = 8 + 18 = 26$ (b) Let $P(n)$ be the proposition $P_n = \frac{2n^3 + 9n^2 + 13n}{3}$. For $n = 1$: L.H.S. $= P_1 = 8$ R.H.S. $= \frac{2(1)^3 + 9(1)^2 + 13(1)}{3} = \frac{24}{3} = 8 = \text{L.H.S.}$ $\therefore P(1)$ is true. Assume $P(m)$ is true for some positive integer m, i.e., $P_m = \frac{2m^3 + 9m^2 + 13m}{3}$ For $n = m + 1$: Note that $\mathbf{v}_{m+1} \cdot \mathbf{v}_{m+2} = (m+1)(m+2) + (m+2)(m+3) = (m+2)(2m+4) = 2m^2 + 8m + 8.$ L.H.S. $= P_{m+1} = P_m + (\mathbf{v}_{m+1} \cdot \mathbf{v}_{m+2})$ $= \frac{2m^3 + 9m^2 + 13m}{3} + (2m^2 + 8m + 8)$ $= \frac{2m^3 + 9m^2 + 13m + 6m^2 + 24m + 24}{3}$ $= \frac{2m^3 + 15m^2 + 37m + 24}{3}$ R.H.S. $= \frac{2(m+1)^3 + 9(m+1)^2 + 13(m+1)}{3}$ $= \frac{2(m^3 + 3m^2 + 3m + 1) + 9(m^2 + 2m + 1) + 13m + 13}{3}$ $= \frac{2m^3 + 15m^2 + 37m + 24}{3} = \text{L.H.S.}$ $\therefore P(m+1)$ is true if $P(m)$ is true. By mathematical induction, $P(n)$ is true for all positive integers n.	1A 1A 1A 1M 1M 1M 1A	